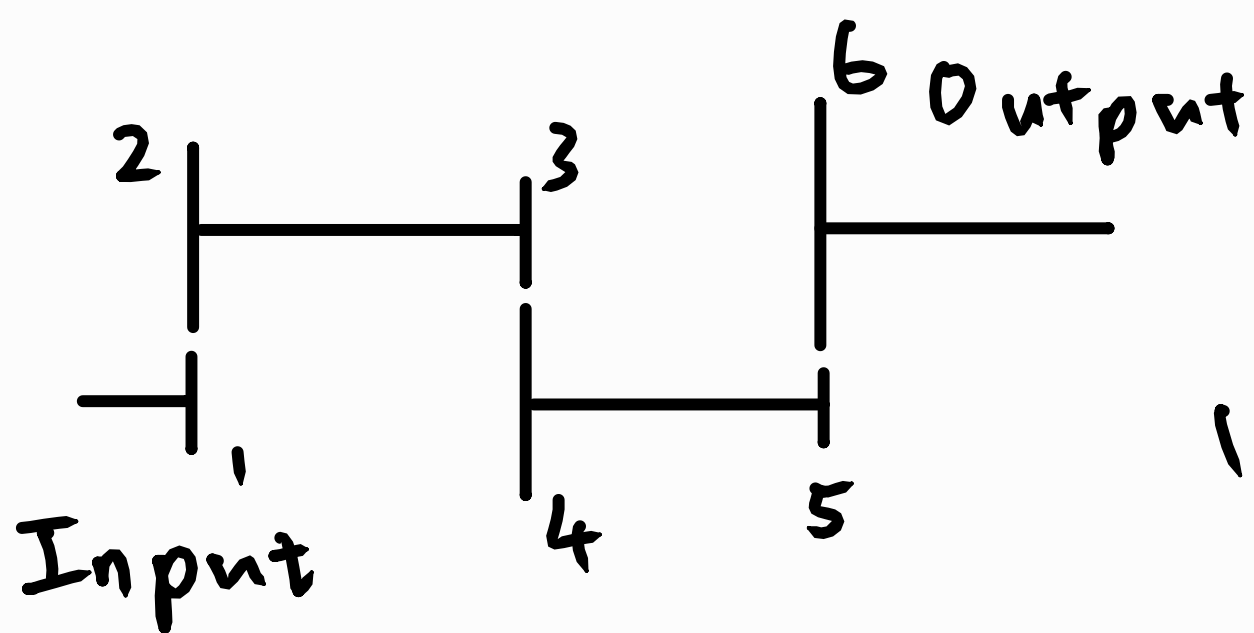




Input
3100 rpm

G
Dr Sim Siang
Kok

130 rpm $\pm 3\%$

$$126.1 \text{ rpm} \leq n_6 \leq 133.9 \text{ rpm}$$

$$N_1 = N_3 = N_5$$

$$N_2 = N_4 = N_6$$

Minimum $N_1 = 15$

$$\text{Speed ratio: } \frac{N_2}{N_1} \times \frac{N_4}{N_3} \times \frac{N_6}{N_5} = \frac{3100}{130}$$

$$\frac{N_2^3}{N_1^3} = \frac{3100}{130}$$

$$\frac{N_2}{N_1} = \sqrt[3]{\frac{3100}{130}}$$

$$= 2.878322464$$

$$\approx 2.88$$

Obtaining the torque on gear 1:

$$P = T\omega$$

$$T_1 = \frac{1.2 \times 10^3}{3100 \times \frac{2\pi}{60}}$$

$$= 3.696501904 \text{ Nm}$$

What is the value
of the torque reading,
 T_c , from the chart?

Is $T_c > T_1$?

Yes

For $N_1 = 15$, looking at graph $z-056$:

$$m_1 = m_2 = 1.25$$

$$\begin{aligned} N_2 &= 2.88 N_1 \\ &= 43.17483696 \\ &\approx 43.17 \end{aligned}$$

From the catalogue $z-105$, $N_2 = 42$,

$$\begin{aligned} T_2 &= \frac{1.2 \times 10^3}{3100 \times \frac{2\pi}{60} \times \frac{15}{42}} \\ &= 10.35020533 \approx 10.35 \text{ Nm} \end{aligned}$$

We use $N_4 = N_2 = 42$ and $N_6 = 45$ to obtain the speed ratio.

Checking the output speed:

$$3100 \times \frac{15}{42} \times \frac{15}{42} \times \frac{15}{45} = 131.8027211 \text{ rpm} \checkmark$$

(within $130 \text{ rpm} \pm 3\%$)

$$T_3 = T_2 = 10.35 \text{ Nm}$$

$$N_3 = N_1 = 15$$

From graph $z-056$:

$$m_3 = m_4 = 1.5$$

$$T_c > T_3 ?$$

Yes.

$$T_4 = \frac{1.2 \times 10^3}{3100 \times \left(\frac{15}{42}\right)^2 \times \frac{2\pi}{60}}$$

$$= 28.98057493$$

$$\approx 28.98 \text{ Nm}$$

$$T_4 = T_5 = 28.98 \text{ Nm}$$

$$N_5 = N_3 = N_1 = 15$$

Is $T_C > T_5$?

Yes.

From graph Z-056,

$$m_5 = m_6 = 2.0$$

$$T_6 = \frac{1.2 \times 10^3}{3100 \times \frac{2\pi}{60} \left(\frac{15}{42}\right)^2 \left(\frac{15}{45}\right)}$$

$$= 86.94172478$$

$$\approx 86.94 \text{ Nm}$$

Chosen gears:

All dimensions in mm

Gear	RPM	Part No.	N	m	d	da	Centre distance
1	3100	SG1215	15	1.25	18.75	21.25	35.625
2	1107	SG1242	42	1.25	52.5	55	
3	1107	SG1515	15	1.5	22.5	25.5	42.75
4	395	SG1542	42	1.5	63	66	
5	395	SG2015	15	2.0	30	34	60
6	132	SG2045	45	2.0	90	94	

$$CD_{1-2} = \frac{18.75 + 52.5}{2}$$

$$= 35.625 \text{ mm}$$

$$CD_{3-4} = \frac{22.5 + 63}{2}$$

$$= 42.75 \text{ mm}$$

$$CD_{5-6} = \frac{30 + 90}{2}$$

$$= 60 \text{ mm}$$

Lowest point is the bottom of gear 4
Highest point is the top of gear 6

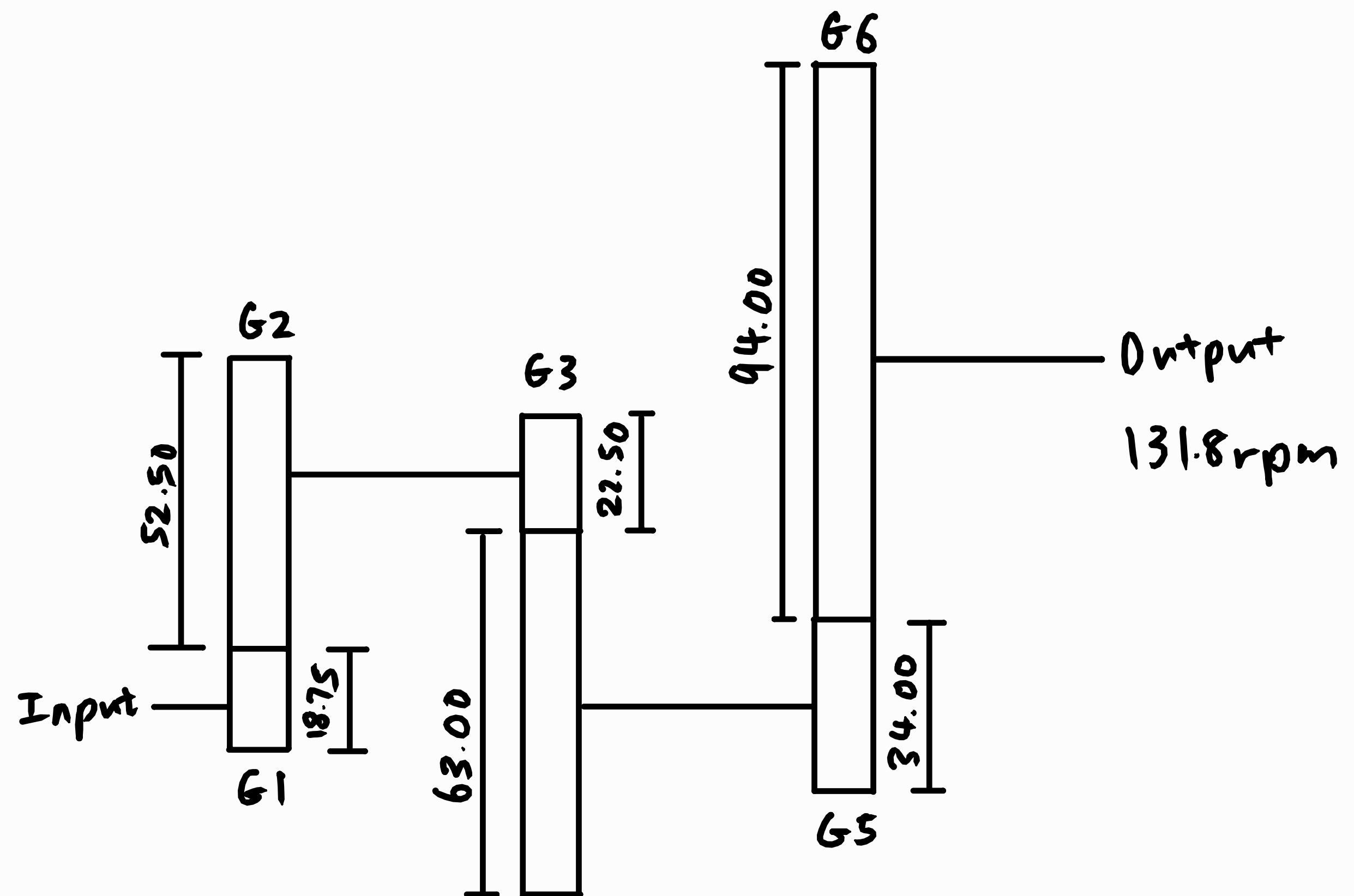
$$\therefore \text{Height} = CD_{5-6} + \frac{da_4}{2} + \frac{da_6}{2}$$

$$= 60 + \frac{66}{2} + \frac{94}{2}$$

$$= 140 \text{ mm} < 175 \text{ mm} \checkmark$$

$$\therefore \text{Width} = CD_{5-6} + \frac{da_5}{2} + \frac{da_6}{2} = 60 + \frac{34}{2} + \frac{94}{2}$$

$$= 124 \text{ mm} < 150 \text{ mm} \checkmark$$



Neat piece of work.

Where is the interference check?

Use:

$$\max N_g = \frac{N_p^2 \sin^2 \phi - 4}{4 - 2N_p \sin^2 \phi}$$

for the max number of teeth on larger gear.

Where is the torque check, $T_c > T_1, T_3, T_5$?

Comment on the final RPM!

Within range.

